

FENIL PATEL

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SUMMARY

M.Sc. in Big Data Analytics from Trent University and B.Sc. in Computer Science, with two years of experience analyzing loan portfolio data. Experienced in building SQL and Python data pipelines, managing PostgreSQL databases, developing machine learning models with scikit-learn and XGBoost, and creating reports and dashboards in Power BI and Tableau. Recent projects include credit default prediction using banking data and locally hosted LLM pipelines for extracting structured information and validating results against ground-truth data

SKILLS

Languages & Databases: SQL (joins, CTEs, window functions, tuning), Python (Pandas, NumPy, scikit-learn, XGBoost), PostgreSQL, Excel (Power Query, PivotTables)

Data Engineering: ETL/ELT pipelines, Prefect orchestration, versioned schema design, append-only modeling, data quality gates, deduplication (SHA-256 hashing), Alembic migrations

ML & AI Systems: Credit risk and classification modelling, feature engineering, imbalanced-data evaluation (PR-AUC, stratified CV), LLM extraction pipelines, local inference (Ollama), constrained outputs (Pydantic), evaluation harness design, prompt versioning, LLM observability (Langfuse)

Analytics & Methods: A/B testing, hypothesis testing (t-test, chi-square, ANOVA), regression and classification, time-series forecasting, anomaly detection, NLP (TF-IDF)

BI & Cloud: Power BI (DAX, star schema, KPIs, RLS), AWS (S3, Glue, Redshift), Git/GitHub

EXPERIENCE

Gatestone & Co. Inc. — Technical Support Representative

Nov 2025 – Jun 2026

- Diagnosed and resolved network, internet, cable, and digital phone issues for customers by phone and electronic channels, working within defined support scope and service-level targets.
- Documented every interaction with a structured log of problem and resolution, building case history other representatives worked from and escalating recurring faults to leadership.
- Promoted product features and upgrades alongside support resolution, meeting monthly incentive targets while holding call quality and handle-time standards.

LoanDash — Data Analyst

Dec 2022 – Dec 2024

- Improved loan approval rates by 18% through portfolio data analysis and partner-bank insights, supporting institutional lending decisions under governance and compliance requirements.
- Built customer segmentation models in Python and SQL that lifted campaign conversions by 15%; mentored teammates on model development and statistical analysis.
- Ran recurring data assessments to surface trend shifts and reconciliation gaps, driving process improvements and data-literacy workshops across teams.

PROJECTS

Berka Credit Risk — Loan Default Prediction on Relational Banking Data · *Python, XGBoost, scikit-learn, SQL, Pandas*

- Engineered 62 features from 7 raw columns across 8 relational tables (accounts, transactions, loans, cards, demographics), assembling the modelling table with SQL joins rather than working from a pre-cleaned extract.
- Trained an XGBoost classifier to 0.926 ROC-AUC and 0.744 PR-AUC under 5-fold stratified cross-validation on an 11% default base rate, reporting PR-AUC as the honest metric for the class imbalance.
- Caught two target-leaking features during feature engineering before they inflated results, and flagged a regional signal as a small-sample artifact (41 loans, chi-square $p = 0.12$) instead of reporting it as a finding.
- Found a plain 180-day minimum balance to be the strongest single predictor, outranking every engineered ratio — documented as evidence for keeping interpretable features in a credit-scoring context.

Complaint Intelligence — LLM Extraction Pipeline on CFPB Consumer Complaints · *Python, Ollama, Qwen 2.5 7B, Pydantic, Pandas*

- Built a locally-hosted LLM pipeline that enriched 10,000 CFPB credit card complaint narratives with 8 structured fields on

a single RTX 4050 (6 GB VRAM) laptop in roughly 10 hours at \$0 API cost — an on-premise pattern suited to regulated data that cannot leave the environment.

- Benchmarked four open-weight models (Qwen 2.5 7B, Qwen 3 8B, Llama 3.1 8B, DeepSeek-R1 7B) on this workload and selected Qwen 2.5 7B for 100% JSON schema conformance at ~4 seconds per extraction; Pydantic-constrained outputs removed malformed generations as a failure class.
- Quantified relief-rate gaps across consumer segments: older consumers received monetary relief on 21.4% of complaints against 6.8% for consumers reporting job loss on comparable complaint types, with one issuer 18.4 points lower on job-loss relief (chi-square $p = 0.026$); reported as descriptive, not causal.

CanadaPathway — Immigration Eligibility & Scoring System · *Python, PostgreSQL, Prefect, Claude API, Pydantic*

- Built a system that ingests Canadian immigration policy from 13 government sources, extracts it into structured rules with an LLM pipeline, and scores user eligibility against a deterministic engine — with every result traceable to the source text it came from.
- Engineered the ingestion layer in Prefect across HTML, RSS, and PDF sources, deduplicating on SHA-256 content hashes and modeling 20+ PostgreSQL tables as append-only history so any past assessment reproduces against the rules live on the day it ran.
- Designed a three-pass extraction architecture (coverage → interpretation → catch-all) that lifted structured field recall from 6 to 39 fields per source, and routed 9 agents across a three-tier model hierarchy to hold full-pipeline inference under \$0.30/day.
- Built an eval harness over 14 canonical profiles with government-cited expected outputs, scoring 14/14 against Canada’s official CRS calculator; anomaly profiling caught a false default inherited by 17 of 19 records that would have silently mis-scored an entire stream.

Bank Loan Portfolio Dashboard — Lending KPI & Risk Reporting · *Power BI, Python, Excel*

- Developed a Power BI dashboard analyzing 38K+ loans (\$435M funded, \$473M received), automating KPI tracking for interest rates, debt-to-income ratios, and non-performing loans (13.8%) to support risk monitoring.
- Created interactive drill-down reporting by purpose, state, and loan grade, enabling portfolio segmentation and opportunity identification for stakeholders.

EDUCATION

Trent University — M.Sc. Big Data Analytics · *GPA 3.9 / 4.0*

Jan 2024 – Apr 2025

Ganpat University — B.Sc. Computer Science · *GPA 3.2 / 4.0*

May 2019 – May 2023

CERTIFICATIONS

IBM Data Science Professional Certificate (Coursera, 2024) · Big Data Foundations with IBM (Coursera, 2024) · Intermediate Python (DataCamp, 2023) · Intermediate SQL (DataCamp, 2023) · AWS Cloud Architecting (AWS Academy, 2023)